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| United States Patent | 12416800               |
| Kind Code            | B2                     |
| Date of Patent       | September 16, 2025     |
| Inventor(s)          | Trivedi; Deepak et al. |

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### Insertion tool

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#### Abstract

An insertion tool for performing an operation on equipment, the insertion tool including: a plurality of segments, each segment of the plurality of segments including a body comprising: a first hinge; and a second hinge, the first hinge of a first segment being coupled to the second hinge of a second segment adjacent to the first segment through an interface, wherein the interface comprises a powder gap, a multi-modal interface, a compliance feature, a displace-to-lock configuration, an interference fit, or any combination thereof, wherein the insertion tool is configured to be selectively rigidizable using a strength member interfacing with the plurality of segments.

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**Appl. No.:** 17/144487

**Filed:** January 08, 2021

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20220221707 A1   | Jul. 14, 2022    |

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#### Publication Classification

**Int. Cl.: G02B23/24** (20060101); **B25J9/06** (20060101); **B25J9/10** (20060101); **B25J18/06** (20060101); **B33Y80/00** (20150101)

**U.S. Cl.:**

**CPC G02B23/2476** (20130101); **B25J18/06** (20130101); B25J9/065 (20130101); B25J9/104 (20130101); B33Y80/00 (20141201); Y10S901/21 (20130101)

## Field of Classification Search

**CPC:** G02B (23/2476); B33Y (80/00); B25J (9/06); B25J (9/1085); B25J (18/06); F05D (2260/36); F05D (2230/80); F01D (25/00); B23P (6/002)

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## **Background/Summary**

### **FIELD**

(1) The present subject matter relates generally to a tool for inspecting an environment and/or performing maintenance operations on a component within the environment, such as within an annular space in a turbine engine.

### **BACKGROUND**

(2) At least certain gas turbine engines include, in serial flow arrangement, a compressor section including a low pressure compressor and a high-pressure compressor for compressing air flowing through the engine, a combustor for mixing fuel with the compressed air such that the mixture may be ignited, and a turbine section including a high pressure turbine and a low pressure turbine for providing power to the compressor section.

(3) Within one or more of the sections, at least certain gas turbine engines define an annular opening. Certain of these annular openings may vary in size, such that a dedicated, specialized insertion tool must be utilized with each annular opening to extend around and through such annular opening.

(4) The inventors of the present disclosure have come up with an insertion tool that may be inserted into an annular opening. The insertion tool that the inventors have come up with may benefit from the inclusion of relatively complex geometries and features. Accordingly, an insertion tool formed in a manner that meets these needs would be useful.

### **BRIEF DESCRIPTION**

(5) Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

(6) In one exemplary embodiment of the present disclosure, an insertion tool for performing an operation on equipment, the insertion tool comprising: a plurality of segments, each segment of the



plurality of segments including a body comprising: a first hinge member; and a second hinge member, the first hinge member of a first segment being coupled to the second hinge member of a second segment adjacent to the first segment through an interface, wherein the interface comprises a powder gap, a multi-modal interface, a compliance feature, a displace-to-lock configuration, an interference fit, or any combination thereof.

(7) According to another exemplary embodiment, an insertion tool for performing an operation on equipment, the insertion tool comprising: a plurality of segments, each segment of the plurality of segments including a body comprising: a first hinge member; and a second hinge member, the first hinge member of a first segment being coupled to the second hinge member of a second segment adjacent to the first segment through an interface, wherein the interface comprises a powder gap, a multi-modal interface, a compliance feature, a displace-to-lock configuration, an interference fit, or any combination thereof; and a strength member intersecting the interface.

(8) According to another exemplary embodiment, a method of forming an insertion tool, the method comprising: additively forming bodies of segments of the insertion tool; and flexing adjacent segments of the insertion tool relative to one another such that powder contained at an interface between adjacent segments of the insertion tool can pass from the interface through a powder gap.

(9) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

(2) FIG. 1 is a cross-sectional schematic view of a high-bypass turbofan jet engine in accordance with an exemplary embodiment of the present disclosure.

(3) FIG. 2 is a close-up, cross-sectional view of a combustion section of the exemplary gas turbine engine of FIG. 1 including an insertion tool in accordance with an exemplary embodiment of the present disclosure, along an axial direction and a radial direction.

(4) FIG. 3 is another close-up, cross-sectional view of the combustion section of the exemplary gas turbine engine of FIG. 1 including the exemplary insertion tool, along the radial direction and a circumferential direction.

(5) FIG. 4 is a perspective view of an insertion tool in accordance with an exemplary embodiment of the present disclosure.

(6) FIG. 5 is a perspective view of an insertion tool in accordance with an exemplary embodiment of the present disclosure.

(7) FIG. 6 is an enlarged view of an end of a segment of an insertion tool in accordance with an exemplary embodiment of the present disclosure.

(8) FIG. 7 is an enlarged view of another end of a segment of an insertion tool in accordance with an exemplary embodiment of the present disclosure.

(9) FIG. 8 is a perspective view of an insertion tool in accordance with an exemplary embodiment of the present disclosure.

(10) FIG. 9 is an enlarged view of a distal end of an insertion tool in accordance with an exemplary embodiment of the present disclosure.

- (11) FIG. 10 is a perspective view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a non-rigidized configuration.
- (12) FIG. 11 is a perspective view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a non-rigidized configuration.
- (13) FIG. 12 is a perspective view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a rigidized configuration.
- (14) FIG. 13A is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a rigidized configuration.
- (15) FIG. 13B is an enlarged schematic view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a rigidized configuration.
- (16) FIG. 14A is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a rigidized configuration.
- (17) FIG. 14B is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a non-rigidized configuration.
- (18) FIG. 15A is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a rigidized configuration.
- (19) FIG. 15B is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a disconnected configuration.
- (20) FIG. 16A is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a rigidized configuration.
- (21) FIG. 16B is a side view of an interface between adjacent segments of an insertion tool in accordance with an exemplary embodiment of the present disclosure as seen in a disconnected configuration.

(22) FIG. 16C is a cross-sectional view of the interface between adjacent segments as seen along Line C-C in FIG. 16A in accordance with an embodiment.

(23) Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

#### DETAILED DESCRIPTION

(24) Reference now will be made in detail to present embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the invention.

(25) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications

and variations as come within the scope of the appended claims and their equivalents.

(26) As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

(27) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust. The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(28) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin.

(29) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(30) In general, an insertion tool in accordance with one or more embodiments described herein can be configured to permit an operator or robotic assembly to service (e.g., inspect and/or repair) a cavity, such as an internal volume of a gas turbine engine. The insertion tool can generally include a plurality of adjacent segments which are selectively rigidizable with respect to one another so as to permit a distal end of the insertion tool access to a confined cavity of the equipment through a complex pathway. Adjacent segments can define hinge members which together form an interface between the adjacent segments. The hinge members can include features to enhance operation of the insertion tool during use, such as when inserting the insertion tool into the internal volume of the gas turbine engine.

(31) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. 1 is a schematic cross-sectional view of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. 1, the gas turbine engine is a high-bypass turbofan jet engine **10**, referred to herein as “turbofan engine **10**.” As shown in FIG. 1, the turbofan engine **10** defines an axial direction A (extending parallel to a longitudinal centerline **12** provided for reference) and a radial direction R. The turbofan engine **10** also defines a circumferential direction C (see FIG. 3) extending circumferentially about the axial direction A. In general, the turbofan **10** includes a fan section **14** and a turbomachine **16** disposed downstream from the fan section **14**.

(32) The exemplary turbomachine **16** depicted is generally enclosed within a substantially tubular outer casing **18** that defines an annular inlet **20** and an annular exhaust **21**. The outer casing **18** encases, in serial flow relationship, a compressor section including a booster or low pressure (LP) compressor **22** and a high pressure (HP) compressor **24**; a combustion section **26**; a turbine section including a high pressure (HP) turbine **28** and a low pressure (LP) turbine **30**; and a jet exhaust nozzle section **32**. A high pressure (HP) shaft or spool **34** drivingly connects the HP turbine **28** to

the HP compressor **24**. A low pressure (LP) shaft or spool **36** drivingly connects the LP turbine **30** to the LP compressor **22**. The compressor section, combustion section **26**, turbine section, and nozzle section **32** together define a core air flowpath **37** therethrough.

(33) For the embodiment depicted, the fan section **14** includes a fixed pitch fan **38** having a plurality of fan blades **40**. The fan blades **40** are each attached to a disk **42**, with the fan blades **40** and disk **42** together rotatable about the longitudinal axis **12** by the LP shaft **36**. For the embodiment depicted, the turbofan engine **10** is a direct drive turbofan engine, such that the LP shaft **36** drives the fan **38** of the fan section **14** directly, without use of a reduction gearbox. However, in other exemplary embodiments of the present disclosure, the fan **38** may instead be a variable pitch fan, and the turbofan engine **10** may include a reduction gearbox, in which case the LP shaft **36** may drive the fan **38** of the fan section **14** across the gearbox.

(34) Referring still to the exemplary embodiment of FIG. **1**, the disk **42** is covered by rotatable front hub **48** aerodynamically contoured to promote an airflow through the plurality of fan blades **40**. Additionally, the exemplary turbofan engine **10** includes an annular nacelle assembly **50** that circumferentially surrounds the fan **38** and/or at least a portion of the turbomachine **16**. For the embodiment depicted, the nacelle assembly **50** is supported relative to the turbomachine **16** by a plurality of circumferentially-spaced outlet guide vanes **52**. Moreover, a downstream section **54** of the nacelle assembly **50** extends over an outer portion of the casing **18** so as to define a bypass airflow passage **56** therebetween. The ratio between a first portion of air through the bypass airflow passage **56** and a second portion of air through the inlet **20** of the turbomachine **16**, and through the core air flowpath **37**, is commonly known as a bypass ratio.

(35) It will be appreciated that although not depicted in FIG. **1**, the turbofan engine **10** may further define a plurality of openings allowing for inspection of various components within the turbomachine **16**. For example, the turbofan engine **10** may define a plurality of borescope openings at various axial positions within the compressor section, combustion section **26**, and turbine section. Additionally, as will be discussed below, the turbofan engine **10** may include one or more igniter ports within, e.g., the combustion section **26** of the turbomachine **16**, that may allow for inspection of the combustion section **26**.

(36) It should further be appreciated that the exemplary turbofan engine **10** depicted in FIG. **1** is by way of example only, and that in other exemplary embodiments, the turbofan engine **10** may have any other suitable configuration, including, for example, any other suitable number of shafts or spools, turbines, compressors, etc. Additionally, or alternatively, in other exemplary embodiments, any other suitable turbine engine may be provided. For example, in other exemplary embodiments, the turbine engine may not be a turbofan engine, and instead may be configured as a turboshaft engine, a turboprop engine, turbojet engine, etc.

(37) Referring now to FIG. **2**, a close-up, schematic view of the combustion section **26** of the turbomachine **16** of the exemplary gas turbine engine **10** of FIG. **1** is provided along with a tool **100** for insertion into an annular section of the engine **10**. It will be appreciated that although the tool **100** is depicted in FIG. **2** as being inserted into a combustion section **26**, in other exemplary embodiments, the tool **100** may additionally, or alternatively, be inserted into other areas of the turbofan engine **10** having an annular shape or other shape. In other embodiments, the tool **100** may be inserted into annular sections of the compressor section or the turbine section, or alternatively still, other engines or systems altogether. For example, FIG. **3** illustrates an embodiment of the tool **100** being inserted into a high pressure (HP) turbine, such as the high pressure turbine **28** previously described. The tool **100** can be inserted into a bore of the engine **10** and passed through the fan section **14**, including fan blades **40**, until reaching an inner portion of the engine **100** corresponding with HP turbine stage-2 C-clips **80**. Certain tool geometry may permit passage of the tool **100** through the high pressure turbine **28** to the desired location of service. The tool **100** can inspect and/or operate on the HP turbine stage 2 C-clips **80** which can be subject to premature failure, resulting in excess aircraft downtime. Additionally or alternatively,

still, the tool **100** may be inserted into a non-annular section. For the embodiment of FIG. 3, the tool **100** capable of insertion into an annular section of an engine is depicted extending through a borescope into the HP turbine **28**.

(38) Referring now also to FIG. 4, providing a partial, axial cross-sectional view of the HP turbine **28** of FIG. 3, it will be appreciated that the tool **100** generally includes a plurality of segments **102** movable into the engine **10**. Each of the plurality of segments **102** can be aligned so as to form a continuous tool **100**. In the rigidized configuration, the plurality of segments **102** can be coupled together such that the tool **100** has a generally rigid structure. That is, the plurality of segments **102** can act like a rigid body exhibiting sufficient structural stiffness so as to maintain a desired shape while moving and/or operating within the engine **10**, e.g., the HP turbine **28**. As illustrated in FIG. 4 and according to certain embodiments, the tool **100** can remain in a one- or two-dimensional spatial arrangement. That is, the tool **100** may not twist, e.g., helically, in a third-dimension of, e.g., a cartesian coordinate system. In other embodiments described herein, the tool **100** may exhibit three-dimensional bending. In one or more embodiments, the tool **100** may be inserted into the engine **10** while having a semi-flaccid configuration. In such a manner, the tool **100** may more readily pass through one or more obstacles in the engine **10**. Once past the obstacles, the tool **100** may be fully rigidized.

(39) In certain instances, the tool **100** can define one or more linear portions **104** and one or more bent portions **106** when in use. Bent portions **106** can be created, for example, at interfaces **108** between adjacent segments **102**. Alternatively, bent portions **106** can be internal to the shape of at least some of the segments **102**. That is, for example, one or more of the segments **102** can define a bent shape that creates a bend in the tool **100** when in the rigidized configuration. The bent portions **106** can define radii of curvature, e.g., R.sub.1. The radius of curvature of the illustrated bent portion **106** can be disposed within a single plane. That is, for example, as described above, the radius of curvature of the bent portion **106** of the tool **100** can be defined by a single plane.

(40) A distal end **110** of the tool **100** can include an implement, which for the embodiment depicted is a camera **112**, to allow for inspection of various components of the high pressure turbine **28**, like the aforementioned C-clips **80** and the like. It will be appreciated, however, that the insertion tool **100** may include any other suitable implement, such that the insertion tool **100** may be utilized for any suitable purpose. For example, the insertion tool **100** may be utilized to inspect the interior of the engine using, e.g., the camera **112**. Additionally, or alternatively, the insertion tool **100** may include various other tool implements to perform one or more maintenance operations within the interior of the engine (e.g., drilling, welding, heating, cooling, cleaning, spraying, etc.).

(41) Further, the exemplary insertion tool **100** can include a drive assembly **114** for driving the insertion tool **100** into, or out of, the interior of the engine, and more specifically for the embodiment shown, into or out of the HT turbine **28**. The drive assembly **114** may be operably coupled to a controller or other control device, such that a length of the insertion tool **100** within the interior of the engine **10** may be controlled with relative precision by the drive assembly **114**. In certain embodiments, the drive assembly **114** can include a motor, servo-motor, or the like configured to drive the tool **100** relative to the engine **10**. In other instances, the drive assembly **114** can include a manual interface configured to permit an operator to manually move the tool **100**. As described hereinafter, the drive assembly **114** can be a selective rigidizer configured to selectively rigidize the tool **100** to a desired shape.

(42) FIG. 5 illustrates a perspective view of the tool **100** as seen in accordance with an exemplary embodiment in the flaccid, e.g., non-rigid, configuration. The tool **100** includes segments **102**, such as a first segment **102A**, a second segment **102B**, a third segment **102C**, a fourth segment **102D**, a fifth segment **102E**, a sixth segment **102F**, a seventh segment **102G**, and an eighth segment **102H**. While the illustrated embodiment depicts the tool **100** as including eight segments **102**, the number of segments **102** may be varied. For instance, the tool **100** can include at least two segments, such as at least three segments, such as at least four segments, such as at least five segments, such as at

least six segments, such as at least seven segments, such as at least eight segments, such as at least nine segments, such as at least ten segments, such as at least fifteen segments, such as at least twenty segments, such as at least thirty segments, such as at least forty segments, and so on. In an embodiment, at least two of the plurality of segments **102** can have same, or similar, shapes as compared to one another. That is, for instance, the at least two segments **102** can have bodies **114** defining same, or similar, sidewalls **116** and/or ends **118** as compared to one another. In a more particular embodiment, all of the plurality of segments **102** can share a common body shape, or a particular aspect of body shape. For instance, all of the plurality of segments **102** can have the same sidewall lengths, as measured between opposing ends **118**, all of the plurality of segments **102** can have a same general sidewall shape, or the like. In another embodiment, at least two of the plurality of segments **102** can have different shapes as compared to one another. For instance, the bodies **114** of at least two segments can have different lengths as compared to one another, different circumferential dimensions as compared to one another, different ends **118**, or the like. By way of non-limiting example, the seventh segment **102G** depicted in FIG. 5 has a length  $L_{sub.7}$  that is less than a length  $L_{sub.6}$  of the sixth segment **102F**. By way of another non-limiting example, the first segment **102A** can be formed of a first material and the second segment **102B** can be formed of a second material different than the first material. By way of yet another non-limiting example, the first segment **102A** can be formed using a particular manufacturing process or manufacturing tolerance different from the manufacturing process or tolerance of the second segment **102B**. For instance, the first segment **102A** can have a lower tolerance, or resolution, than the second segment **102B**. In certain instances, the segments **102** can be formed using an additive manufacturing process, such as three-dimensional printing. The first segment **102A** can have a lower print resolution as compared to the print resolution of the second segment **102B**, the third segment **102C**, and so on. This may occur, for example, where the first segment **102A** is a handle or outermost segment which does not require high surface finish characteristics for sliding over or past features of the engine **10** during navigation therethrough.

(43) In one or more embodiments, each pair of adjoining, i.e., neighboring, segments **102** can be attached together through an interface **120**. The interface **120** may be disposed at, or adjacent to, ends **118** of the adjacent segments **102**. Referring, for example, to the interface **120** between the fourth and fifth segments **102D** and **102E**, each interface **120** can be formed from a first hinge member **122** associated with one of the segments, e.g., the fourth segment **102D**, and a second hinge member **124** associated with the adjacent segment, e.g., the fifth segment **102E**. The first and second hinge members **122** and **124** can be joined together to permit relative movement between the adjacent segments **102**, e.g., between the fourth segment **102D** and the fifth segment **102E**. By way of example, the first and second hinge members **122** and **124** can permit relative movement, e.g., rotational movement, of the segments **102** in one or more planes. In a particular embodiment, the interface **120** between a pair of adjacent segments **102** can permit movement of the segments **102** in a single plane. For instance, as illustrated in FIG. 5, the fourth and fifth segments **102D** and **102E** can be moveable with respect to one another along a plane corresponding with arrow **126**. That is, the fourth and fifth segments **102D** and **102E** can pivot relative to one another along the directions shown by arrow **126** while staying within a single plane of relative motion.

(44) In the illustrated embodiment, the interfaces **120** formed between at least two pairs of adjacent segments **102** can be different from one another. For example, the interface **120** formed between the fourth and fifth segments **102D** and **102E** is disposed in a first plane of rotation while the interface **120** formed between the second and third segments **102B** and **102C** is disposed in a second plane of rotation different from the first plane. Accordingly, the angle of rotation of the interface **120** formed between the third and fourth segments **102C** and **102D** can be different from the angle of rotation of the interface **120** formed between the fourth and fifth segments **102D** and **102E**. In the non-rigid configuration, as illustrated for example in FIG. 5, such multi-planar interfacing may not materially affect the flaccid shape of the tool **100**. However, when rigidized,

the multi-planar interfacing depicted in FIG. 5 can result in a tool **100** having a three-dimensional shape for accessing certain areas of the engine **10** or similar structure being serviced (inspected and/or operated upon).

(45) A distal segment of the tool **100**, such as the eighth segment **102H** in the depicted embodiment, can have a dissimilar shape as compared to the other segments **102** for purpose of permitting servicing operations. In the illustrated embodiment, the eighth segment **102H** is depicted as having a tapered profile with a minimum width disposed at or adjacent to a distal end **110** of the tool **100**. In such a manner, the tool **100** may be more readily fed into the equipment being serviced, e.g., the aircraft engine. Moreover, the tapered profile may permit the implement, e.g., camera **112**, to exit an internal volume of the tool **100** (described in greater detail below) so as to perform an operation during the service without requiring the diameter of the tool **100** to change.

(46) FIGS. 6 and 7 illustrate ends **118** of adjoining, i.e., neighboring, segments **102**. The ends **118** can be matched to one another such that the interface **120** (FIG. 5) therebetween moves in a predetermined manner, e.g., in a predetermined plane as compared to other interfaces **120** formed between other pairs of segments **102**.

(47) In an embodiment, the body **114** of at least one of the segments **102** can be formed through an additive manufacturing process, such as by way of non-limiting example, can include three-dimensional printing. Bodies **114** in accordance with some embodiments described herein can thus include indicia of the three-dimensional printing manufacturing process in the form of indicia, including stratum, e.g., layers, formed in the body **114** corresponding with individually stepped printing layers. In certain embodiments described herein, all segments **102** of the tool **100** can be formed using additive manufacturing processes, e.g., three-dimensional printing techniques. In a particular embodiment, the segments **102** can be additively manufactured simultaneously while already in the interfaced configuration. That is, adjacent segments **102** can be additively formed in engaged position relative to one another.

(48) Referring initially to FIG. 6, the exemplary segment **102** depicted may refer to any one or more of the aforementioned segments **102** (e.g., the first segment **102A**, the second segment **102B**, the third segment **102C**, and so on). The body **114** of the segment **102** defines the first hinge member **122** of the interface **120** (FIG. 5). The first hinge member **122** is laterally offset from a central axis **128** of the body **114** in a radial direction. That is, the first hinge member **122** may not be centrally disposed with respect to the body **114**. In accordance with the particular embodiment depicted in FIG. 6, the first hinge member **122** generally includes a central structure **130** and a channel **132** disposed adjacent thereto. The channel **132** is a split channel, including a first channel portion **132A** and a second channel portion **132B**. The central structure **130** is disposed between the first and second channel portions **132A** and **132B**. In an embodiment, the first and second channel portions **132A** and **132B** can have the same, or similar, shapes and/or sizes as compared to one another. In another embodiment, the first and second channel portions **132A** and **132B** can have different shapes and/or sizes as compared to one another. The central structure **130** can define opposite surfaces **134A** and **134B** spaced apart from one another by a thickness of the central structure **130**. The opposite surfaces **134A** and **134B** can form end walls of the first and second channel portions **132A** and **132B**. As described below with respect to FIG. 7, the opposite surfaces **134A** and **134B** can be engaged with complementary surfaces of the second hinge member **124** to form the interface **120** between the adjacent segments **102**.

(49) The body **114** of the segment **102** illustrated in FIG. 6 further includes a cavity **136**. The cavity **136** extends through the length of the segment **102** and can emerge from the body **114** at two or more exit locations, such as at exit location **138**. In the illustrated embodiment, the exit location **138** is shown intersecting the first hinge member **122**. That is, for example, the exit location **138** can emerge from the body **114** at an exit location transverse, or generally transverse, to an axis of rotation of the interface **120** through the first hinge member **122**. A second exit location (not illustrated) of the cavity **136** can exit the body **114** of the segment **102** through the a second hinge

member of the segment **102**. In such a manner, at least one of the bodies **114** can include both first and second hinge members **122** and **124** and the cavity **136** can exit the body **114** through the first and second hinge members **122** and **124**. The cavity **136** can define a constant, or generally constant, cross-sectional shape along the length of the body **114**. In certain instances, the cavity **136** can be linear, or generally linear. That is, a longitudinal axis of the cavity **136** can lie along a straight, or generally straight line. As described in greater detail below, the cavity **136** can be configured to receive a strength member. The strength member may form a backbone of the tool **100** offset from the central axis **128** of the segments **102** and thus offset from the central axis of the tool **100**. In some embodiments, the strength member can comprise a flexible member, such as, e.g., a tension bearing element, string, memory-laden material defining a predefined shape, or the like. The strength member can allow for flexure of the tool **100** while permitting the tool **100** to return to, e.g., a predefined shape. For example, the tool **100**, in a non-rigid configuration, can slide through a predefined volume of an aircraft engine until reaching a desired location. The tool **100** may have to undergo distortion, e.g., bending, to navigate the predefined volume of the aircraft engine. For instance, the tool **100** may have to slide around corners and through shaped passageways to reach the desired location. The strength member can permit the tool to remain in a single, operable piece for sliding into the airplane engine while preventing the tool from becoming jammed or stopped within the predefined volume. The strength member can put the tool **100** in the predefined shape once the tool is positioned at the desired location to permit operation on the engine.

(50) In an embodiment, the strength member can occupy less than an entire areal dimension of the cavity **136**. For example, the strength member can be a hollow tube extending through the cavity **136**. By way of another example, the strength member can have a cross-sectional shape different from the cavity **136** and/or a size that is smaller than the cavity **136**. In such a manner, the cavity **136** can further define space to receive the implement, such as the aforementioned camera **112** therethrough.

(51) In an embodiment, the strength member may include a shape memory alloy (SMA) material. In a more particular embodiment, the strength member can be formed entirely from an SMA material. In yet another particular embodiment, the strength member can be at least partially formed from an SMA material. An SMA is generally an alloy capable of returning to its original shape after being deformed. Further, SMAs may act as a lightweight, solid-state alternative to traditional materials. For instance, certain SMAs may be heated in order to return a deformed SMA to its pre-deformed shape. An SMA may also provide varying stiffness, in a predetermined manner, in response to certain ranges of temperatures. The change in stiffness of the shape memory alloy is due to a temperature related, solid state micro-structural phase change that enables the alloy to change from one physical shape to another physical shape. The changes in stiffness of the SMA may be developed by working and annealing a preform of the alloy at or above a temperature at which the solid state micro-structural phase change of the shape memory alloy occurs. The temperature at which such phase change occurs is generally referred to as the critical temperature or transition temperature of the alloy.

(52) Some shape memory alloys used herein are characterized by a temperature-dependent phase change. These phases include a martensite phase and an austenite phase. The martensite phase generally refers to a lower temperature phase whereas the austenite phase generally refers to a higher temperature phase. The martensite phase is generally more deformable, while the austenite phase is generally less deformable. When the shape memory alloy is in the martensite phase and is heated to above a certain temperature, the shape memory alloy begins to change into the austenite phase. The temperature at which this phenomenon starts is referred to as the austenite start temperature ( $A_s$ ). The temperature at which this phenomenon is completed is called the austenite finish temperature ( $A_f$ ). When the shape memory alloy, which is in the austenite phase, is cooled, it begins to transform into the martensite phase. The temperature at which this transformation starts is



referred to as the martensite start temperature (Ms). The temperature at which the transformation to martensite phase is completed is called the martensite finish temperature (Mf). As used herein, the term “transition temperature” without any further qualifiers may refer to any of the martensite transition temperature and austenite transition temperature. Further, “below transition temperature” without the qualifier of “start temperature” or “finish temperature” generally refers to the temperature that is lower than the martensite finish temperature, and the “above transition temperature” without the qualifier of “start temperature” or “finish temperature” generally refers to the temperature that is greater than the austenite finish temperature.

(53) In some embodiments, the strength member has a first stiffness at a first temperature and has a second stiffness at a second temperature, wherein the second temperature is different from the first temperature. Further, in some embodiments, one of the first temperature and the second temperature is below the transition temperature and the other one may be at or above the transition temperature. Thus, in some embodiments, the first temperature may be below the transition temperature and the second temperature may be at or above the transition temperature, while in some other embodiments, the first temperature may be at or above the transition temperature and the second temperature may be below the transition temperature.

(54) Exemplary but non-limiting examples of SMAs may include nickel-titanium (NiTi) and other nickel-titanium based alloys such as nickel-titanium hydrogen fluoride (NiTiHf) and nickel-titanium palladium (NiTiPd). However, it should be appreciated that other SMA materials may be equally applicable to the current disclosure. For instance, in certain embodiments, the SMA may include a nickel-aluminum based alloys, copper-aluminum-nickel alloy, or alloys containing zinc, copper, gold, and/or iron. The alloy composition may be selected to provide the desired stiffness effect for the application such as, but not limited to, damping ability, transformation temperature and strain, the strain hysteresis, yield strength (of martensite and austenite phases), resistance to oxidation and hot corrosion, ability to change shape through repeated cycles, capability to exhibit one-way or two-way shape memory effect, and/or a number of other engineering design criteria. Suitable shape memory alloy compositions that may be employed with the embodiments of present disclosure may include, but are not limited to NiTi, NiTiHf, NiTiPt, NiTiPd, NiTiCu, NiTiNb, NiTiVd, TiNb, CuAlBe, CuZnAl and some ferrous based alloys. In some embodiments, NiTi alloys having transition temperatures between 5° C. and 150° C. are used. NiTi alloys may change from austenite to martensite upon cooling.

(55) The body **114** further includes one or more auxiliary cavities **140**. In an exemplary embodiment, the auxiliary cavities **140** may be disposed on an opposite side of the central axis **128** as compared to the cavity **136**. The one or more auxiliary cavities **140** can include, for example, at least one auxiliary cavity, such as at least two auxiliary cavities, such as at least three auxiliary cavities. In an embodiment, the auxiliary cavities **140** may have one or more same, or similar attributes as compared to one another. For instance, the auxiliary cavities **140** can all share a same radial offset distance from the central axis **128**. In another embodiment, the auxiliary cavities **140** may have one or more different attributes as compared to one another. For instance, the auxiliary cavities **140** can have different diameters as compared to one another. As described in greater detail hereinafter, in certain embodiments the auxiliary cavities **140** may be configured to receive one or more support members. In certain instances, one or more selectively rigidizable element(s), e.g., support member(s) and/or strength member, may operate to selectively rigidize the tool **100** and/or help support the tool **100** in the non-rigidized, i.e., flaccid, configuration. For example, where the support members comprise tension bearing materials, e.g., a string, tensioning the support members can result in the tool **100** taking the rigidized configuration. During tensioning of the support members, adjacent segments of the tool **100** may pivot relative to, e.g., around, interfaces **120** until the support members reach a critical tension whereby the tool is rigidized. In certain instances, the strength member may also be selectively rigidizable by applying tension thereto (e.g., FIG. **11**). In other embodiments, the selectively rigidizable elements can be selectively rigidizable using a non-

tensioning method. For example, selective rigidization can occur through a phase shift, chemical and/or electrical stimulation, and the like.

(56) A hole **142** can be disposed within the body **114** of at least one of the segments **102**, such as all of the segments **102**, and configured to receive an implement, such as the aforementioned camera **112**, for performing an operation at the desired location. In an embodiment, the hole **142** can be centrally, or generally centrally, located relative to the central axis **128** of the body **114**. In certain instances, the hole **142** can be disposed at a radial position between the cavity **138** and at least one of the auxiliary cavities **140**. The hole **142** can define a non-circular cross section. For instance, the hole **142** can define an ovular cross-sectional profile, a rectangular cross sectional profile, or another shape other than a circle. In certain instances, the hole **142** can be configured to receive an implement, such as a cable connected to a tool, e.g., camera **112**, that has a non-circular cross section. By way of example, the cable can be a ribbon cable or another flat, or generally flat, cable. The cable can be configured to bend in a single, or generally single, axis. That is, for instance, the cable can define a planar shape. The planar shape can bend in a direction perpendicular to the planar shape. The hole **142** can be shaped and/or oriented relative to the direction of movement at the interface **120** (FIG. 5) such that the planar cable bends in a direction associated with a direction of movement, e.g., rotation, at the interface **120**. In an embodiment, the hole **142** can define an aspect ratio [W.sub.H:H.sub.H], as defined by a maximum relative width, W.sub.H, of the hole **142**, as compared to a maximum relative height, H.sub.H, of the hole **142**, that is at least 1.5:1, such as at least 2:1, such as at least 3:1, such as at least 4:1, such as at least 5:1, such as at least 7.5:1, such as at least 10:1, such as at least 15:1, such as at least 20:1. In a particular embodiment, the width and height of the hole **142** can be oriented generally perpendicular with respect to one another.

(57) FIG. 7 illustrates an exemplary embodiment of a segment **102** disposed adjacent to the segment **102** illustrated in FIG. 6. The segment **102** illustrated in FIG. 7 includes end **118** having the second hinge member **124** which is configured to engage with the first hinge member **122** described with respect to the segment **102** illustrated in FIG. 6. The second hinge member **124** can be complementary in shape and/or size with respect to the first hinge member **122** so as to permit engagement between the adjacent segments **102** illustrated in FIGS. 6 and 7. Accordingly, the second hinge member **124** can include a central structure **144** and a ridge **146**. The central structure **144** can be disposed centrally along the ridge **146**. The central structure **144** can split the ridge **146** into a first ridge portion **146A** and a second ridge portion **146B**. The central structure **144** can be indented into the ridge **146** so as to receive the central structure **130** of the first hinge **120**. The central structure **144** can define opposite surfaces **144A** and **144B** configured to couple with surfaces **134A** and **134B** of the first hinge **120** so as to form the interface **120**.

(58) The body **114** of the segment **102** can further define a cavity **148** configured to receive the strength member exiting the cavity **138** of the adjacent segment **102**. The cavity **148** can extend through the length of the segment **102** and emerge from the body **114** at two or more exit locations, such as at exit location **150**. In the illustrated embodiment, the exit location **150** is shown intersecting the second hinge member **124**. That is, for example, the exit location **150** can emerge from the body **114** at an exit location transverse, or generally transverse, to an axis of rotation of the interface **120** through the second hinge member **124**. A second exit location (not illustrated) of the cavity **148** can exit the body **114** of the segment **102** through the first hinge member **122** of the segment **102**. In such a manner, the cavity **148** can exit the body **114** through the first and second hinge members **122** and **124**. The cavity **148** can define a constant, or generally constant, cross-sectional shape along the length of the body **114**. In certain instances, the cavity **148** can be linear, or generally linear. That is, a longitudinal axis of the cavity **148** can lie along a straight, or generally straight line.

(59) The body **114** further includes one or more auxiliary cavities **152**. In an exemplary embodiment, the auxiliary cavities **152** may be disposed on an opposite side of a central axis **154** as compared to the cavity **148**. The one or more auxiliary cavities **152** can include, for example, at

least one auxiliary cavity, such as at least two auxiliary cavities, such as at least three auxiliary cavities. In an embodiment, the auxiliary cavities **152** may have one or more same, or similar attributes as compared to one another. For instance, the auxiliary cavities **152** can all share a same radial offset distance from the central axis **154**. In another embodiment, the auxiliary cavities **152** may have one or more different attributes as compared to one another. For instance, the auxiliary cavities **152** can have different diameters as compared to one another. The auxiliary cavities **152** may be configured to receive the aforementioned one or more support members.

(60) A hole **156** can be disposed within the body **114** of the segment **102** and configured to receive the aforementioned implement, e.g., cable, extending through the hole **142** of the adjacent segment **102**. In an embodiment, the hole **156** can be centrally, or generally centrally, located relative to the central axis **154** of the body **114**. In certain instances, the hole **156** can be disposed at a radial position between the cavity **148** and at least one of the auxiliary cavities **152**. The hole **156** can define a non-circular cross section. For instance, the hole **156** can define an ovular cross-sectional profile, a rectangular cross sectional profile, or another shape other than a circle. In certain instances, the hole **156** can be configured to receive an implement, such as a cable connected to a tool, e.g., camera **112**, that has a non-circular cross section. By way of example, the cable can be a ribbon cable or another flat, or generally flat, cable. The cable can be configured to bend in a single, or generally single, axis. That is, for instance, the cable can define a planar shape. The planar shape can bend in a direction perpendicular to the planar shape. The hole **156** can be shaped and/or oriented relative to the direction of movement at the interface **120** (FIG. 5) such that the planar cable bends in a direction associated with a direction of movement, e.g., rotation, at the interface **120**. In an embodiment, the hole **156** can define an aspect ratio  $[W_{sub.H}:H_{sub.H}]$ , as defined by a maximum relative width,  $W_{sub.H}$ , of the hole **156**, as compared to a maximum relative height,  $H_{sub.H}$ , of the hole **156**, that is at least 1.5:1, such as at least 2:1, such as at least 3:1, such as at least 4:1, such as at least 5:1, such as at least 7.5:1, such as at least 10:1, such as at least 15:1, such as at least 20:1. In a particular embodiment, the width and height of the hole **156** can be oriented generally perpendicular with respect to one another.

(61) While the above description refers separately to the cavities **136** and **150**, the auxiliary cavities **140** and **152**, and holes **142** and **156**, it should be understood that these aspects can share any common geometry and/or shape as compared to one another. Specifically, the cavities, auxiliary cavities, and holes may be configured to operate together to perform the above-described functions for the tool. As such, these features may be, but are not required to be, common to all segments **102** of the tool **100**. Accordingly, reference to particular aspects with respect to one but not all of these elements as it relates to one segment **102** may refer to that same aspect pertaining to all segments **102**. Additionally, in another embodiment the cavities **136** and **150** may be part of the same cavity, each auxiliary cavity **140** and **152** can be part of the same auxiliary cavity, and holes **142** and **156** can be part of the same hole. That is, FIG. 6 and FIG. 7 can illustrate different, e.g., opposite, ends of the same segment **102**.

(62) FIG. 8 illustrates an embodiment of the tool **100** in the rigidized configuration. The tool **100** is contained within box **158**. As seen in FIG. 8, according to some embodiments of the present disclosure, the tool **100** can define a complex geometry extending through a three dimensional cartesian coordinate system. That is, the tool **100** may simultaneously extend in the X-, Y-, and Z-axis. The particular shape of the tool **100** can be viewed in response to the shape of the environment in which the tool is to be used within.

(63) The tool **100** can further include an elongated portion **160** shown in box **162**. The elongated portion **160** may be connected with the tool **100** to elongate the tool **100** for insertion into certain engine components. The elongated portion **160** may be coupled to the tool **100** at a joint **164**. The joint **164** may be a removable joint, such that an operator can install the tool **100** on a plurality of different elongated portions **160** of variable size, shape, or configuration. In certain embodiments, the elongated portion **160** can include a multi-piece construction. For instance, the elongated

portion **160** may include a first elongated segment **166** and a second elongated segment **168** coupled together. Accordingly, the operator can select various segments of various shaped, sized, and/or configurations based on the specific application or requirement of the intended use. A handle **170** may be coupled to a proximal end **172** of the elongated portion **160** to permit an operator to hold the tool **100** at a desired location. The handle **170** may include an interface **174** to be engaged with the elongated portion **160**. In a non-illustrated embodiment, the handle **170** may be directly coupled with the tool **100**. That is, the elongated portion **160** may be omitted in accordance with one or more embodiments.

(64) The handle **170** may include one or more elements configured to permit selective rigidization of the tool **100**. For example, the handle **170** can include a trigger **176** configured to selectively rigidize the tool **100**. The trigger **176** can be rotatable, pivotal, translatable, or the like between a disengaged configuration in which the tool **100** is flaccid, i.e., non-rigid, and an engaged configuration in which the tool is rigid. In a particular embodiment, the trigger **176** may include finger grips (not illustrated) which permit an operator to maintain positive contact with the trigger **176** when pushing and pulling the trigger between the engaged and disengaged positions. In another particular embodiment, the trigger **176** may include multi-point contact locations, e.g., two discrete finger grip locations. In an embodiment, the trigger **176** may be selectively lockable in the engaged and/or disengaged positions. Accordingly, the operator can rigidize the tool **100** and release the trigger **176** during servicing operations. In yet a further embodiment, the trigger **176** can be operated by a motor or other power device such that rigidization of the tool **100** does not require the generation of manual pressure. The handle **170** may remain exposed from the engine while the tool **100** is inserted in the engine. In such a manner, the operator can maintain control of the tool **100** without direct access thereto.

(65) The handle **170** can include an orifice **178** or other interface configured to receive an implement for insertion into the engine. The orifice **178** can include, for example, an opening into which the operator can insert the implement, e.g., the aforementioned camera **112** on a cable, into the engine. In certain embodiments, the orifice **178** may be selectively sealable or closable to prevent ingress of dust and other contaminants from entering the tool **100**. Once opened, the operator can feed and/or move the implement into or relative to the orifice **178** so as to allow for the servicing operation to be performed as required.

(66) FIG. **9** illustrates an embodiment of the distal end **110** of the tool **100**. The final segment **102** of the tool **100** disposed at the distal end **110** can include an opening **180** (which may be similar to or the same as the cavity **136** and/or **148**) in which the camera **112** can extend through. In such a manner, the camera **112** can monitor a leading edge of the tool **100** as it enters the engine and as it performs servicing operations including inspection of the engine and/or repair operations. As previously described, the camera **112** can be disposed within a circumferential boundary of the tool **100**. That is, in accordance with an embodiment, the camera **112** can be disposed within an outer diameter of the segments **102** such that the camera does not create a larger leading edge diameter as compared to the diameter of the segments **102**. The aforementioned holes **142** and **156** can be in communication with a final hole **182** through which an implement, such as a flat cable, can be passed through. As described above, the implement can be hand, or machine, fed through the orifice **178** (FIG. **8**) to the distal end **110** for performing the servicing operation.

(67) FIGS. **10** and **11** illustrate an embodiment of an insertion tool **1000** with a strength member **1002** extending through cavities **1004** of adjacent segments **1006**. The strength member **1002** illustrated in FIGS. **10** and **11** includes a plurality of strength members, such as at least two strength members, such as at least three strength members. However, the strength member **1002** may alternatively include a single strength member. The insertion tool **1000** is illustrated in the non-rigidized configuration wherein the segments **1006** are not coupled together at a fixed angular disposition. That is, the adjacent segments **1006** can readily move relative to each other. FIG. **12** illustrates the insertion tool **1000** with segments **1006** fixedly coupled together by the strength

member **1002**. Selectively rigidizing the tool **1000**, i.e., transitioning from the non-rigidized configuration illustrated in FIGS. **10** and **11**, to the rigidized configuration illustrated in FIG. **12**, can occur, for example, by tensioning the strength member **1002** and/or one or more support members **1016** disposed opposite the cavities **1004**. As tension is applied, the length of the strength member **1002** and/or support members **1016** within the tool **1000** can decrease, effectively pulling adjacent segments **1006** together in the direction shown by arrows **1008** (FIG. **11**). As the segments **1006** come into contact with one another, internal tension within the strength member **1002** and/or support members **1016** increases, thereby maintaining the segments **1006** connected together, and effectively rigidizing the tool **1000**. Ends **1010** of the segments **1006** can be keyed together to prevent relative movement in the rigidized configuration. Keyed ends **1010** can include complementary face characteristics that, when combined, prevent relative movement, e.g., rotation, between the adjacent segments **1006**. By way of non-limiting example, the keyed ends **1010** can include complementary posts and recesses, complementary undulating features, complementary castellations, and the like. As the complementary features are mated with one another, the adjacent segments **1006** can become fixed relative to one another and maintained at their fixed relative position by tension within the strength member **1002**.

(68) In the illustrated embodiment, the adjacent pair of depicted segments **1006** include complementary ridges **1012** and channels **1014** configured to align with one another in the rigidized configuration. In addition to securing the segments **1006** at relatively fixed positions with respect to one another, the ridges **1012** and channels **1014** can act as hinges for the segments **1006** when the strength member **1002** is less than fully tensioned. That is, for instance, when tension on the strength member **1002** is lessened, the segments **1006** may deflect relative to one another. If tension is maintained above a certain amount, such that the ridges **1012** and channels **1014** do not fully unseat from one another, the segments **1006** may move within a guided track relative to each other about a pivot axis formed by the ridges **1012** and channels **1014**. This may be advantageous during insertion of the tool **1000** into the engine **10** as the tool **1000** may be forced to navigate complex geometry within the engine which requires the tool **1000** to bend while simultaneously retaining some amount of rigidity.

(69) As previously described, selective rigidization of the tool **1000** can occur through use of one or more support members **1016**. The one or more support members **1016** can extend through auxiliary cavities **1018** of the segments **1006**. The support members **1016** can help retain the segments **1006** together, for instance, when the strength member **1002** is in the non-fully tensed state. The support members **1016** can further assist in guiding the tool **1000** within the engine **10** during the tool insertion process. The support members **1016** can also be used to selectively rigidize the tool **1000** while the strength member **1002** forms a flexible backbone for the tool **1000**.

(70) As described with respect to FIGS. **6** and **7**, adjacent segments **102** of the tool may be joined together at interface **120** by one or more hinge members, e.g., a first hinge member **122** and a second hinge member **124**. The first and second hinge members **122** and **124** can act together to connect the segments **102** while permitting relative movement therebetween, particularly when the tool **100** is not rigidized. The first and second hinge members **122** and **124** described with respect to FIGS. **6** and **7** are held together through interference fit. In particular, surfaces **134A** and **134B** engage with surfaces **144A** and **144B**, respectively, to dynamically hold adjacent segments **102** together. FIGS. **13** to **18** depict alternate embodiments of the interface **120** and associated hinges **122** and **124**.

(71) FIG. **13A** illustrates an enlarged view of an interface **120** formed between adjacent segments **102** in accordance with an exemplary embodiment. FIG. **13B** illustrates an enlarged, cross-sectional view of the interface as seen along Line A-A in FIG. **13A**. Referring initially to FIG. **13A**, the segments **102** can define first and second hinge members **122** and **124** as described above. The first and second hinge members **122** and **124** can cooperatively mate with one another to permit selectively flexure between the segments **102**.

(72) In one or more embodiments, the first and second hinge members **122** and **124** can be integrally formed with the body **114** of the segment **102**. In a particular embodiment, the segments **102**, including the first and second hinge members **122** and **124**, can be formed using an additive manufacturing process, such as for example, three-dimensional printing, resulting in a stratum including a plurality of layers. In yet a more particular embodiment, the segments **102** can be formed in the assembled state such that adjacent segments **102** are, e.g., additively manufactured with the resulting interfaces **120** formed during the manufacturing process and ready for use. While additively manufacturing the interfaces **120** with the first and second hinge members **122** and **124** assembled may reduce the introduction of weakness into the bodies **114** resulting from flexing the bodies to install the first and second hinge members **122** and **124** relative to one another, certain additive manufacturing processes can also result in the formation of excess material left at the interface **120**. For instance, certain three-dimensional printing processes utilize printing powder which is formed into the final shape of the structure being printed. Not all of the printing powder may solidify, resulting in powder being caught within the interface **120**. Such stray powder can reduce efficiency of the tool **100**, potentially resulting in premature wear and/or failure while the tool **100** is disposed within the engine **10** or even result in powder being deposited in the engine as the tool **100** navigates therethrough during insertion therein.

(73) To mitigate the confluence of powder within the interface **120**, the first and second hinge members **122** and **124** can be shaped so as to have a powder gap **184** configured to permit powder to pass therethrough. In particular, the powder gap **184** may permit passage of powder through the powder gap **184** when the tool **100** is flexed or otherwise deformed. Referring to FIG. **13B**, the powder gap **184** may be formed by a gap between the first and second hinge members **122** and **124**. Powder **186** can exit the interface **120** through the powder gap **184** in a direction corresponding, for example, with arrow **188**. To further enhance powder removal from the interface **120**, in certain instances at least one of the first and second hinge members **122** and **124** can have a tapered interface profile. For instance, an inner sidewall **190** of the second hinge member **124** can have an outward tapered profile. The inner sidewall **190** can define a taper angle,  $\beta$ , as measured with respect to a normal axis **194**, of at least one degree, such as at least two degrees, such as at least three degrees, such as at least four degrees, such as at least five degrees, such as at least six degrees, such as at least seven degrees, such as at least eight degrees, such as at least nine degrees, such as at least ten degrees, such as at least fifteen degrees. In another embodiment, the outer sidewall **192** of the first hinge member **122** can have an inward tapered profile. The outer sidewall **194** can define a taper angle,  $\alpha$ , as measured with respect to the normal axis **194**, of at least one degree, such as at least two degrees, such as at least three degrees, such as at least four degrees, such as at least five degrees, such as at least six degrees, such as at least seven degrees, such as at least eight degrees, such as at least nine degrees, such as at least ten degrees, such as at least fifteen degrees. In certain instances,  $\alpha$  can be equal, or approximately equal, with  $\beta$ . In other instances,  $\alpha$  and  $\beta$  can be different from one another. In an embodiment, the inner sidewall **190** and the outer sidewall **192** can be angularly offset from one another.

(74) FIG. **14A** illustrates an enlarged view of an interface **120** formed between adjacent segments **102** in accordance with another exemplary embodiment when the tool **100** is in a rigid, or semi-rigid, configuration. FIG. **14B** illustrates an enlarged view of the interface **120** of FIG. **14A** when the tool is in a non-rigid configuration, i.e., the segments **102** are not rigidly held together and have moved relative to one another. The interface **120** depicted in FIGS. **14A** and **14B** is multi-modal. Multi-modal interfacing can include instance of close-fit and other instances of loose fit. In the rigid configuration, the first and second hinge members **122** and **124** have a close-fitting arrangement whereby accurate alignment between the segments **102** occurs. In the non-rigid configuration, the first and second hinge members **122** and **124** can have a loose fit whereby the interface **120** is relaxed. Close-fitting arrangement can be the result of hinge geometry. For instance, the first hinge member **122** can define an elongated length,  $L_{sub.1}$ , and the second hinge

member **124** can define an elongated length, L.sub.2. In the rigid configuration, L.sub.1 and L.sub.2 can be offset from one another. For instance, L.sub.1 and L.sub.2 can be rotationally offset from one another by at least one degree, such as at least two degrees, such as at least three degrees, such as at least four degrees, such as at least five degrees, such as at least six degrees, such as at least seven degrees, such as at least eight degrees, such as at least nine degrees, such as at least ten degrees, such as at least fifteen degrees, such as at least twenty degrees, such as at least thirty degrees, such as at least forty degrees, such as at least fifty degrees, such as at least sixty degrees, such as at least seventy degrees, such as at least eighty degrees, such as at least ninety degrees. In the non-rigid configuration, L.sub.1 and L.sub.2 can be offset from one another by an amount less than in the rigid configuration. By way of non-limiting example, in a particular embodiment L.sub.1 and L.sub.2 can be offset from one another by ninety degrees in the rigid configuration and offset by less than ninety degrees in the non-rigid configuration.

(75) In an embodiment, the first hinge member **122** comprises a post **210** and the second hinge member **124** comprises a recess **212** into which the post **210** is insertable. At least one of the post **210** and recess **212** can have an aspect ratio different than 1:1, wherein the aspect ratio is defined by a length of the post **210** and/or recess **212**, e.g., L.sub.1, relative to a width thereof, as measured perpendicular to the length. That is, for example, at least one of the first and second hinge members **122** and **124** can have an elongated shape.

(76) FIG. 15A illustrates an enlarged view of an interface **120** formed between adjacent segments **102** in accordance with another exemplary embodiment. FIG. 15B illustrates an enlarged view of the interface **120** between adjacent segments **102** as seen in a disconnected configuration for purpose of viewing the first and second hinge members **122** and **124**. In an embodiment, the first and second hinge members **122** and **124** can be configured to be connected to one another when the adjacent pair of segments **102** are disposed at a first orientation with respect to one another. After adjusting the adjacent pair of segments **102** to a second orientation different from the first orientation, the adjacent pair of segments **102** can remain coupled together. This type of engagement may be described as displaced-to-lock engagement. By way of example, the displacement can include angular displacement. The angular displacement between the first and second orientations can be at least 1°, such as at least 2°, such as at least 3°, such as at least 4°, such as at least 5°, such as at least 10°, such as at least 20°, such as at least 30°, such as at least 40°, such as at least 50°, such as at least 60°, such as at least 70°, such as at least 80°. The displacement can also include translation, pivoting, the like, or any combination thereof.

(77) In the embodiment illustrated in FIG. 15B, the first hinge member **122** includes receiving structure **196** comprising a first side surface **198** and a second side surface **200**. The first and second side surfaces **198** and **200** can extend from at least one of opposite side surfaces **134A** and **134B** of the first hinge member **122**. The first and second side surfaces **198** and **200** define a gap **202** that extends therebetween. The second hinge member **124** includes a projection **204** configured to be passed through the gap **202** when the segments are oriented relative to one another at the first orientation and retained between the first and second side surfaces **198** and **200** when the segments **102** are oriented relative to one another at the second orientation. A retaining element **206** can additionally be passed through the gap **202** when the first and second hinge members **122** and **124** are joined together in the first orientation. In the connected (i.e., engaged) configuration resulting from reorienting the segments **102** to the second orientation, the retaining element **206** can interface with the first side surface **198** to further support and maintain engagement between the segments **102**. In an embodiment, the receiving structure **196** can be approximately the same on opposite side surfaces **134A** and **134B** of the first hinge member **122**. In another embodiment, the projection **204** and retaining element **206** can be approximately the same on both opposite side surfaces **144A** and **144B** of the second hinge member **124**.

(78) FIG. 16A illustrates an enlarged view of an interface **120** formed between adjacent segments **102** in accordance with another exemplary embodiment. FIG. 16B illustrates an enlarged view of

the interface **120** between adjacent segments **102** as seen in a disconnected configuration for purpose of viewing the first and second hinge members. FIG. **16C** illustrates a cross-sectional view of the interface **120** between adjacent segments **102** as seen along Line C-C in FIG. **16A**. The interface **120** includes a compliance feature **208** disposed between opposite side surfaces **144A** and **144B** of the second hinge member **124**. The compliance feature **208** can permit expansion between the opposite side surfaces **144A** and **144B** during assembly of the segments **102**. In an embodiment the compliance feature **208** can include a slit extending into the body **114** of the segment. The slit can define a length of at least 0.1 mm, such as at least 0.2 mm, such as at least 0.3 mm, such as at least 0.4 mm, such as at least 0.5 mm, such as at least 1 mm, such as at least 2 mm, such as at least 3 mm, such as at least 4 mm, such as at least 5 mm. As the first and second hinge members **122** and **124** are brought together so as to be engaged with one another, a diameter of the slit can increase, allowing the opposite surfaces **134A** and **134B** to pass over and onto the opposite side surfaces **144A** and **144B**. In an embodiment, the opposite surfaces **134A** and **134B** of the first hinge member **122** can be complementary to opposite side surface **144A** and **144B** of the second hinge member **124**. Complementary side surfaces can include, for example, flat surfaces, conical surface, hemispherical surfaces, and the like which fit together so as to maintain the first and second hinge members **122** and **124** coupled together at the interface **120**.

(79) In accordance with one or more embodiments described herein, the first and second hinge members **122** and **124** can be configured to float relative to one another. As used herein, “float relative to one another” refers to an interface whereby two or more members, such as the first and second hinge members **12** and **124** of adjacent segments **102**, are not fixedly coupled together and can move relative to one another, e.g., in the relaxed state. Instead, the adjacent segments **102** are held together, for example, by the strength member and/or support member, i.e., in the selectively rigidized state. Conversely, in certain instances, when tension is removed, the adjacent segments **102** may separate from one another when the first and second hinge members **122** and **124** float relative to one another.

(80) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

(81) Further aspects of the invention are provided by the subject matter of the following clauses:

(82) Embodiment 1. An insertion tool for performing an operation on equipment, the insertion tool comprising: a plurality of segments, each segment of the plurality of segments including a body comprising: a first hinge member; and a second hinge member, the first hinge member of a first segment being coupled to the second hinge member of a second segment adjacent to the first segment through an interface, wherein the interface comprises a powder gap, a multi-modal interface, a compliance feature, a displace-to-lock configuration, an interference fit, or any combination thereof.

(83) Embodiment 2. The insertion tool of any one or more of the embodiments, wherein the body of at least one of the segments comprises a stratum comprising a plurality of layers.

(84) Embodiment 3. The insertion tool of any one or more of the embodiments, wherein the insertion tool comprises a strength member transversely intersecting the interface.

(85) Embodiment 4. The insertion tool of any one or more of the embodiments, wherein the powder gap is disposed between an outer sidewall of the first hinge member and an inner sidewall of the second hinge member.

(86) Embodiment 5. The insertion tool of any one or more of the embodiments, wherein the inner and outer sidewalls are angularly offset from one another.



(87) Embodiment 6. The insertion tool of any one or more of the embodiments, wherein the multi-modal interface comprises a close fit when the adjacent segments are rigidized by the strength member and a loose fit when the adjacent segments are not rigidized by the strength member.

(88) Embodiment 7. The insertion tool of any one or more of the embodiments, wherein the first hinge member comprises a post and the second hinge member comprises a recess into which the post is insertable, and wherein at least one of the post and recess have an aspect ratio different than 1:1.

(89) Embodiment 8. The insertion tool of any one or more of the embodiments, wherein the first hinge member comprises a first outer surface and a second outer surface, wherein the second hinge member comprises a first inner surface and a second inner surface, and wherein the interface comprises a hemispherical interface between the first outer surface and the first inner surface and a hemispherical interface between the second outer surface and the second inner surface.

(90) Embodiment 9. The insertion tool of any one or more of the embodiments, wherein the displace-to-lock interface comprises a projection of the second hinge member configured to pass through a gap between adjacent side surfaces of a receiving structure of the first hinge member.

(91) Embodiment 10. An insertion tool for performing an operation on equipment, the insertion tool comprising: a plurality of segments, each segment of the plurality of segments including a body comprising: a first hinge member; and a second hinge member, the first hinge member of a first segment being coupled to the second hinge member of a second segment adjacent to the first segment through an interface, wherein the interface comprises a powder gap, a multi-modal interface, a compliance feature, a displace-to-lock configuration, an interference fit, or any combination thereof; and a strength member intersecting the interface.

(92) Embodiment 11. The insertion tool of any one or more of the embodiments, wherein the strength member comprises a backbone of the insertion tool configured to maintain the plurality of segments coupled together when the insertion tool is not rigidized.

(93) Embodiment 12. A method of forming an insertion tool, the method comprising: additively forming bodies of segments of the insertion tool; and flexing adjacent segments of the insertion tool relative to one another such that powder contained at an interface between adjacent segments of the insertion tool can pass from the interface through a powder gap.

(94) Embodiment 13. The method of any one or more of the embodiments, wherein flexing adjacent segments of the insertion tool is performed by tensioning a strength member extending between the adjacent segments.

(95) Embodiment 14. The method of any one or more of the embodiments, wherein the adjacent segments include a first segment having a first hinge member and a second segment having a second hinge member, and wherein the first and second hinge members define the interface between the first and second segments.

(96) Embodiment 15. The method of any one or more of the embodiments, wherein the first and second hinge members are configured to move relative to one another when the tool is not rigidized.

(97) Embodiment 16. The method of any one or more of the embodiments, wherein the interface further comprises a multi-modal interface, a compliance feature, a displace-to-lock configuration, an interference fit, or any combination thereof.

(98) Embodiment 17. The method of any one or more of the embodiments, wherein additively forming the bodies comprises three-dimensional printing.

(99) Embodiment 18. The method of any one or more of the embodiments, further comprising rigidizing the insertion tool by applying force to the strength member.

(100) Embodiment 19. The method of any one or more of the embodiments, wherein the adjacent segments are configured to rotate by an engagement displacement between rigidized and non-rigidized configurations, and wherein flexing adjacent segments of the insertion tool relative to one another such that powder contained at the interface between adjacent segments of the insertion tool

can pass from the interface through the powder gap is performed by rotating the adjacent segments by a rotational displacement no greater than the engagement displacement.

(101) Embodiment 20. The method of any one or more of the embodiments, further comprising coupling the adjacent segments together after additively manufacturing the bodies of the segments.

## Claims

1. An insertion tool for performing an operation on equipment, the insertion tool comprising: a plurality of segments including adjacent first and second segments connected to one another via an interface; wherein the interface comprises a first hinge member of the first segment and a second hinge member of the second segment; wherein the first hinge member is coupled to the second hinge member; wherein the first hinge member comprises a post and the second hinge member comprises a recess in which the post is received; wherein the post and recess allow the first and second segments to pivot relative to one another; wherein the first and second segments are shiftable between a non-rigid configuration, wherein the post is positioned in the recess with a loose fit, and a rigid configuration, wherein the post is positioned in the recess with a close-fit; wherein the post and recess are configured to be shifted between the loose fit and close-fit by rotating at least one of the first and second segments with respect to one another about an axis of rotation extending through the post and the recess while the post is retained in the recess; and wherein the post and the recess each have an elongate configuration such that the post and the recess each have an aspect ratio different than 1:1 such that when at least one of the first and second segments is rotated with respect to one another about the axis of rotation extending through post and the recess, the post and recess are shifted from the loose fit to the close-fit, wherein the post contacts the recess at opposing outer surface portions of the post, wherein said aspect ratio is defined as a length and width of a cross section of an end face of the post and a cross section of the recess; wherein the first and second hinge members define the axis of rotation about which the first and second segments are rotatable, the axis of rotation being transverse to a longitudinal axis of each of the first and second segments; wherein the post extends from a surface of the first segment in a direction parallel to the axis of rotation.
2. The insertion tool of claim 1, wherein a body of at least one of the plurality of segments comprises a stratum comprising a plurality of sequentially-formed layers.
3. The insertion tool of claim 1, wherein the first segment comprises a third hinge member and the second segment comprises a fourth hinge member, wherein the third hinge member is positioned in the fourth hinge member to allow the first and second segments to pivot relative to one another in cooperation with the first and second hinge members.
4. The insertion tool of claim 1, wherein the post and recess are configured to allow the first and second segments to rotate only about the axis of rotation when the first and second segments are shifted from the non-rigid configuration to the rigid configuration such that the first and second segments have a predefined orientation in the rigid configuration.
5. The insertion tool of claim 1, wherein the post and the recess are each laterally offset in a radial direction from the longitudinal axis of the respective first and second segments such that a gap is formed between ends of the first and second segments that increases in size when the first and second segments are pivoted away from one another about the axis of rotation.
6. The insertion tool of claim 1, wherein the insertion tool comprises an elongate strength member extending between the first and second segments and transversely intersecting the interface, wherein the strength member is configured to shift the first and second segments between the non-rigid configuration to the rigid configuration.
7. The insertion tool of claim 6, wherein the interface comprises the close-fit when the first and second segments are rigidized by the strength member and the loose fit when the first and second segments are not rigidized by the strength member.

8. The insertion tool of claim 1, wherein the interface comprises a powder gap disposed between an outer surface of the post and an inner facing surface of the recess.
  9. The insertion tool of claim 8, wherein the inner facing surface of the recess and the outer surface of the post are angularly offset from one another.
  10. An insertion tool for performing an operation on equipment, the insertion tool comprising: a plurality of segments including a first segment and an adjacent second segment connected to one another via an interface; wherein the interface comprises a first hinge member of the first segment and a second hinge member of the second segment; wherein the first hinge member is coupled to the second hinge member; wherein the first hinge member comprises a post and the second hinge member comprises a recess in which the post is received, wherein the recess comprises an endless inner facing surface that surrounds the post; wherein the post and recess allow the first and second segments to pivot relative to one another; wherein the first and second segments are shiftable between a non-rigid configuration, wherein the post is positioned in the recess with a loose fit, and a rigid configuration, wherein the post is positioned in the recess with a close-fit; a strength member extending between the first and second segments and intersecting the interface, wherein the strength member is configured to shift the first and second segments between the non-rigid configuration and the rigid configuration; wherein the post and recess are configured to be shifted between the loose fit and close-fit by rotating at least one of the first and second segments with respect to one another about an axis of rotation extending through the post and the recess while the post is retained in the recess; and wherein the post and the recess each have an aspect ratio different than 1:1 such that when at least one of the first and second segments is rotated with respect to one another about the axis of rotation extending through the post and the recess, the post and recess are shifted from the loose fit to the close-fit, wherein said aspect ratio is defined as a length and width of a cross section of the post and the recess; wherein the first and second hinge members define the axis of rotation about which the first and second segments are rotatable, the axis of rotation being transverse to a longitudinal axis of each of the first and second segments; wherein the post extends from a surface of the first segment in a direction parallel to the axis of rotation.
  11. The insertion tool of claim 10, wherein the strength member comprises a backbone of the insertion tool configured to maintain the plurality of segments coupled together when the insertion tool is not rigidized.
  12. The insertion tool of claim 10, wherein a body of at least one of the plurality of segments comprises a stratum comprising a plurality of sequentially-formed layers.
  13. The insertion tool of claim 10, wherein the first and second hinge members are configured to move relative to one another when the tool is not rigidized.
  14. The insertion tool of claim 10, wherein at least one of the post and recess include a tapered interface profile.
  15. The insertion tool of claim 10, wherein the interface comprises a powder gap disposed between an outer surface of the post and the endless inner facing surface of the recess.
  16. The insertion tool of claim 15, wherein the endless inner facing surface of the recess and the outer surface of the post are angularly offset from one another.
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